

CHAPTER-3

Statically Indeterminate Structures

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Objectives

At the end of this lesson students are able to

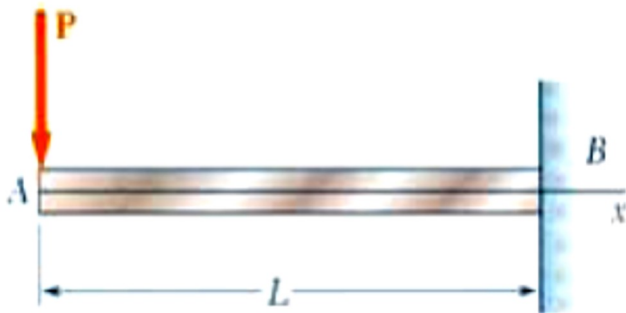
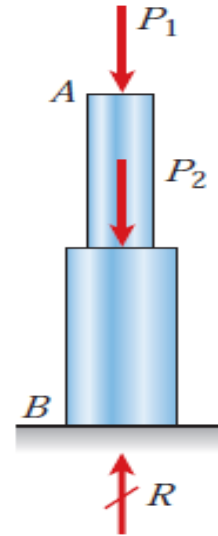
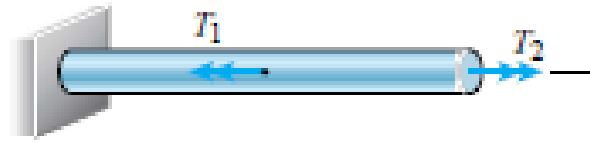
- Identify statically determinate and indeterminate structures and their degree of statically indeterminacy.
- Solve statically indeterminate axial members, torsional members and beams using different methods

3.1 Introduction

➤ **Statically determinate structures** : These are structures in which their reactions and internal forces can be determined solely from free-body diagrams and equations of equilibrium.

Example:

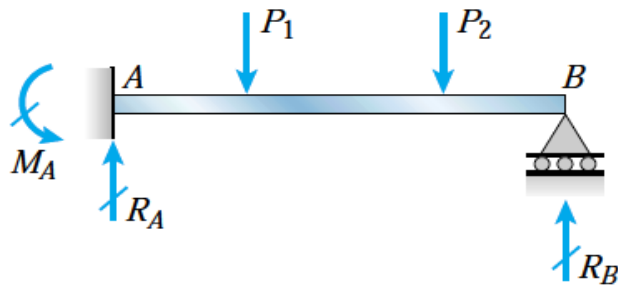
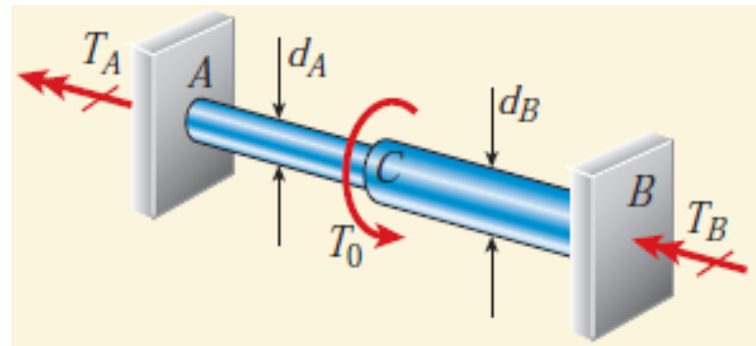
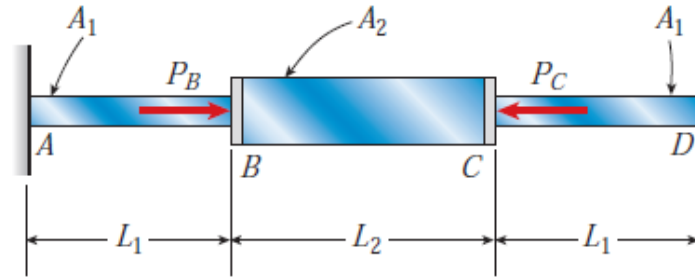
- Axially Loaded Bars
- Torsional Shafts
- Beams



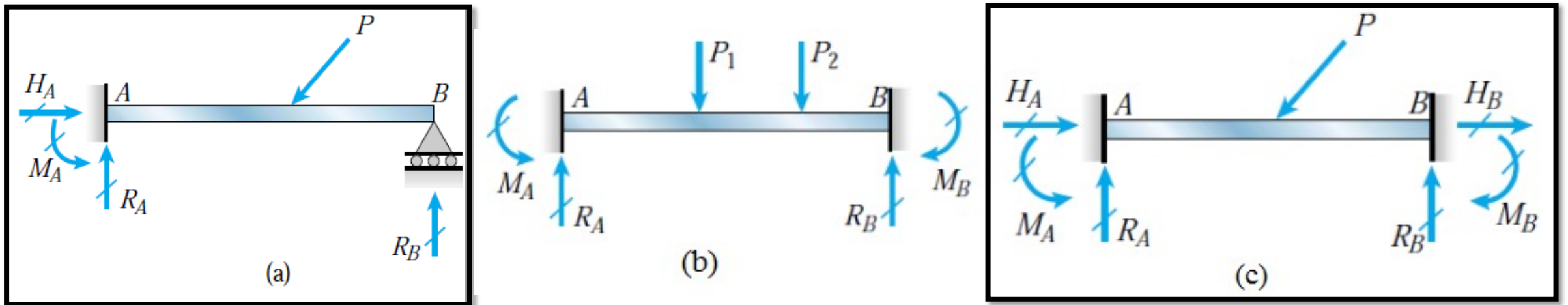
➤ **Statically Indeterminate structures:** These are structures in which their reactions and internal forces cannot be found by statics alone.

Example:

- Axially Loaded Bars
- Torsional Shafts
- Beams



- Any type of structure is classified as ***statically indeterminate*** if the number of ***unknown reactions*** exceeds the available number of ***equilibrium equations*** (No. of unknown. Reactions > No. of equilibrium. Eqns.)
- The additional support reactions that are not needed to keep the structure in stable equilibrium are called ***redundants***.
- The number of these redundants is referred to as ***degree of indeterminacy***



Statically Indeterminate beams of (a) 1st degree (b) 2nd degree (c) 3rd degree

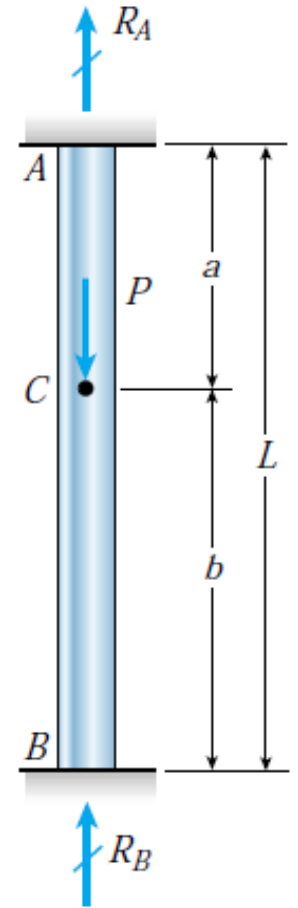
Method of solving Statically Indeterminate Structures

- The analysis of a statically indeterminate structure involves setting up and solving:
 - i) **Equilibrium Equations** - Relate the *loads* acting on the structure to the *unknown forces* (which may be reactions or internal forces)
 - ii) **Compatibility Equations** - Express conditions on the *displacements* of the structure
 - iii) **Force-displacement relations** - Expressions that use the dimensions and properties of the structural members to relate the *forces* and *displacements* of those members.

- For the relatively simple structures, the above method of analysis is adequate.
- However, more formalized approaches are needed for complicated structures.
- There are two commonly used methods which are used in the analysis of ***complicated*** statically indeterminate structures.
 - 1) **Superposition Method** (*Force method or Flexibility method*)
 - 2) **Displacement method** (*Stiffness method*)

3.2 Statically Indeterminate Axial Members

- Consider an axial prismatic bar AB which is attached to rigid supports at both ends and is axially loaded by a force P at an intermediate point C .
- The reactions R_A and R_B cannot be found by statics alone, thus this is a type of statically indeterminate problem of *first degree*
- The reactions at the supports can be calculated as follows



(a) Equilibrium Equation:

- Using the free body diagram the equilibrium equation is expressed as

$$\sum F_{vert} = 0, R_A - P + R_B = 0 \dots\dots\dots (i)$$

(b) Compatibility Equation:

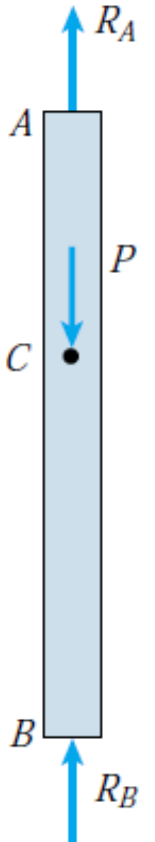
- Since the bar is fixed on both sides, the total deformation is zero

$$\delta_{AB} = 0 \dots\dots\dots (ii)$$

(c) Force-displacement relations:

- The deformations of segment AC and BC due to their corresponding internal forces is

$$\delta_{AC} = \frac{R_A a}{EA} \quad \delta_{CB} = -\frac{R_B b}{EA} \dots\dots\dots (iii)$$



- By combining the force-displacement relations with the equation of compatibility

$$\delta_{AB} = \delta_{AC} + \delta_{CB} = \frac{R_A a}{EA} - \frac{R_B b}{EA} = 0$$

..... (iv)

➤
$$R_A = \frac{b}{a} R_B$$

- Substituting eqn.(iv) in eqn.(i), the reactions R_A and R_B obtained as

$$R_A = \frac{b}{L} P$$

$$R_B = \frac{a}{L} P$$

Example 3.1:

1. As shown in Fig. 3.1a, a rigid horizontal bar is supported by a hinge at A and by two steel cables BD and CE, which are of equal length, $L = 0.8$ m, and cross-sectional area, $A = 140$ mm². Calculate the stress in each cable due to a force of 40 kN, applied as shown in the figure. Assume that the yielding stress is 250 MPa and that $E = 200$ Gpa.

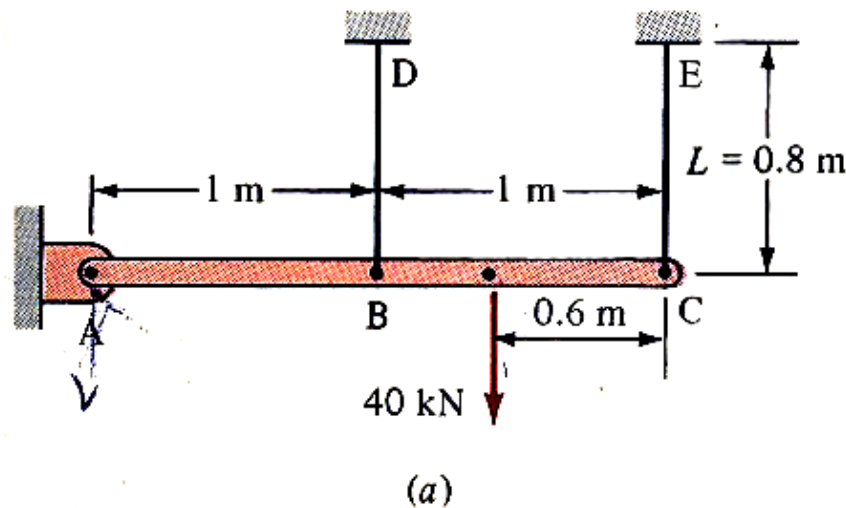
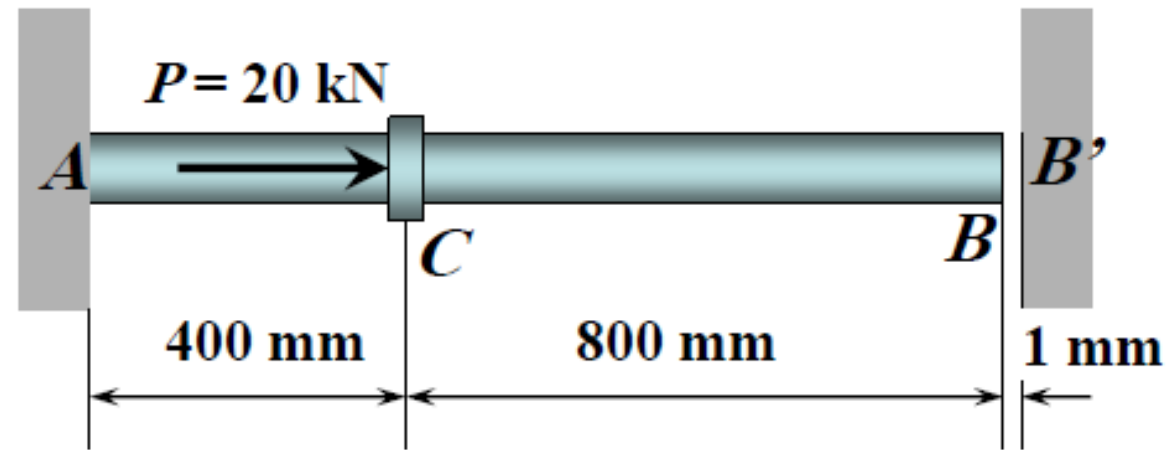


Fig 3.1

Example 3.2:

2. The steel rod shown in the fig. has a diameter of 5 mm. It is fixed to the wall at A, and before it is loaded, there is a gap of 1 mm between the wall at B' and the rod.

Determine the reactions at A and B' if the rod is subjected to axial force of $P = 20 \text{ kN}$ as shown. Neglect the size of the collar at C. Take $E = 200 \text{ GPa}$.

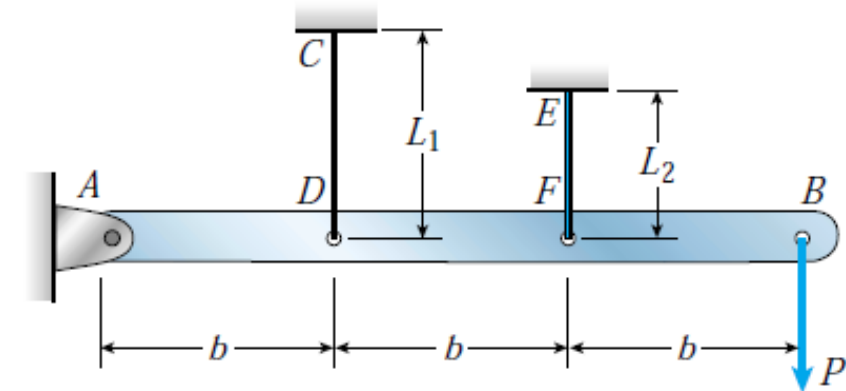


Example 3.3:

3. A horizontal rigid bar AB is pinned at end A and supported by two wires (CD and EF) at points D and F (See Fig.). A vertical load P acts at end B of the bar. The bar has length $3b$ and wires CD and EF have lengths L_1 and L_2 , respectively. Also, wire CD has diameter d_1 and modulus of elasticity E_1 ; wire EF has diameter d_2 and modulus E_2 .

a) Obtain formulas for the allowable load P if the allowable stresses in wires CD and EF , respectively, are σ_1 and σ_2 . (Disregard the weight of the bar itself.

b) Calculate the allowable load P for the following conditions: Wire CD is made of aluminium with modulus $E_1 = 72$ GPa, diameter $d_1 = 4.0$ mm, and length $L_1 = 0.4$ m. Wire EF is made of magnesium with modulus $E_2 = 45$ GPa, diameter $d_2 = 3.0$ mm, and length $L_2 = 0.30$ m. The allowable stresses in the aluminium and magnesium wires are $\sigma_1 = 200$ MPa and $\sigma_2 = 175$ MPa, respectively.

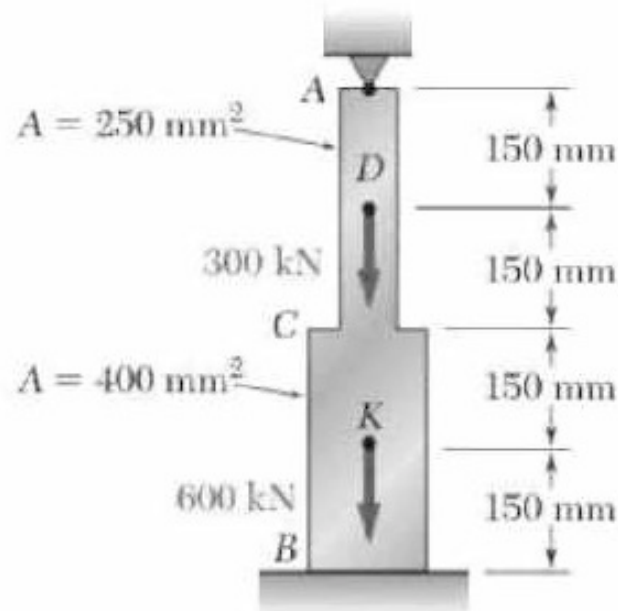


Method of Super Position (Force or Flexibility Method)

It is also possible to solve statically indeterminate problems by writing the compatibility equation using the principle of superposition

Example 3.4:

4. Using method of super position, determine the reactions at A and B for the steel bar and loading shown in Fig., assuming a close fit at both supports before the loads are applied



3.3 Statically Indeterminate torsional members

- Statically Indeterminate torsional members can be analysed by supplementing the *equilibrium equations* with *compatibility equations* pertaining to the rotational displacements.
- Thus, the general method for analysing statically indeterminate torsional members is the same as statically indeterminate bars with axial loads.

- The steps used include :

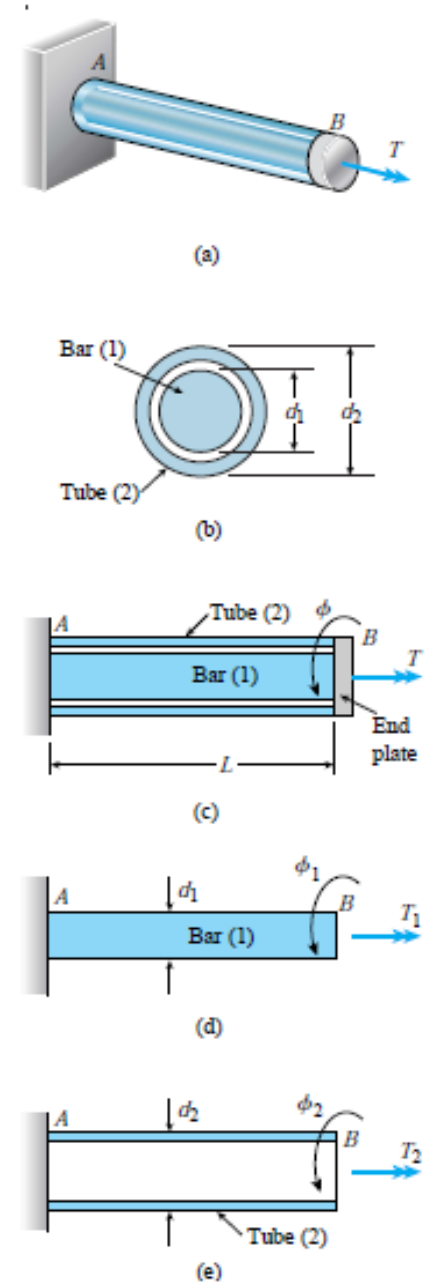
Step 1: write **equations of equilibrium**, obtained from free-body diagrams of the given physical situation

Step 2: write **equations of compatibility**, based upon physical conditions pertaining to the angles of twist

Step 3: relate the angles of twist to the torques by **torque-displacement relations**,

$$\phi = \frac{TL}{GJ} \dots\dots\dots (i)$$

- Consider a composite bar AB shown in Fig.(a). The bar is attached to a fixed support at end A and loaded by a torque T at end B .
- Furthermore, the bar consists of two parts: a solid bar and a tube (Figs.(b) and (c), with both the solid bar and the tube joined to a rigid end plate at B .
- When the torque T is applied to the composite bar, the end plate rotates through a small angle Φ (Fig. (c) and torques T_1 and T_2 are developed in the solid bar and the tube, respectively (Figs. (d) and (e)).
- From equilibrium we know that the sum of these torques equals the applied torque, and so the *equation of equilibrium* is



(a) Equilibrium Equation:

- From equilibrium we know that the sum of these torques equals the applied load, and so the *equation of equilibrium* is

$$T_1 + T_2 = T \dots\dots\dots \text{(ii)}$$

(b) Compatibility Equation:

- Considering the rotational displacements ϕ_1 and ϕ_2 of both the solid bar and the tube respectively. These angles of twist must be equal because the bar and tube are securely joined to the end plate and rotate with it

$$\phi_1 = \phi_2 \dots\dots\dots \text{(iii)}$$

(c) Force-displacement relations:

- The angles ϕ_1 and ϕ_2 are related to the torques T_1 and T_2 by the *torque-displacement relations*, which in the case of linearly elastic materials are obtained from the eqn.(i) as

$$\phi_1 = \frac{T_1 L}{G_1 J_1} \quad \phi_2 = \frac{T_2 L}{G_2 J_2} \quad \dots\dots\dots \text{(iv)}$$

- When the preceding expressions for ϕ_1 and ϕ_2 are substituted into Eq. (iii), the equation of compatibility becomes

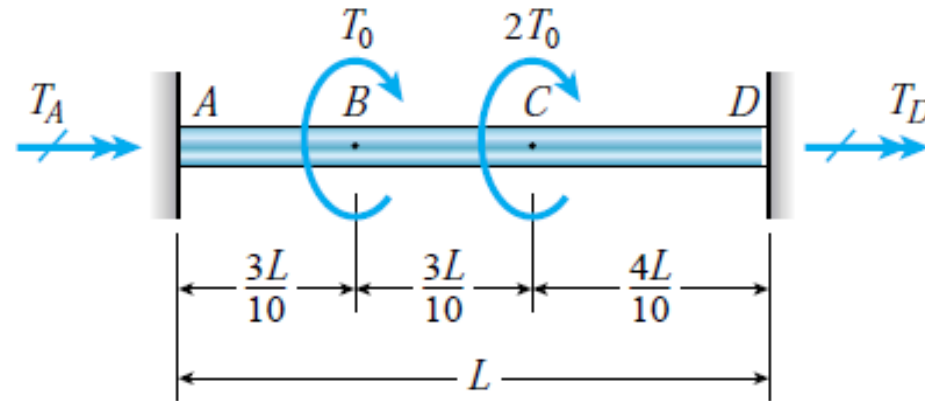
$$\frac{T_1 L}{G_1 J_1} = \frac{T_2 L}{G_2 J_2} \quad \dots\dots\dots \text{(v)}$$
$$T_1 = \frac{G_1 J_1}{G_2 J_2} T_2$$

- Substituting eqn.(v) in eqn.(ii), the torques T_1 and T_2 obtained as

$$T_1 = T \left(\frac{G_1 J_1}{G_1 J_1 + G_2 J_2} \right) \quad T_2 = T \left(\frac{G_2 J_2}{G_1 J_1 + G_2 J_2} \right)$$

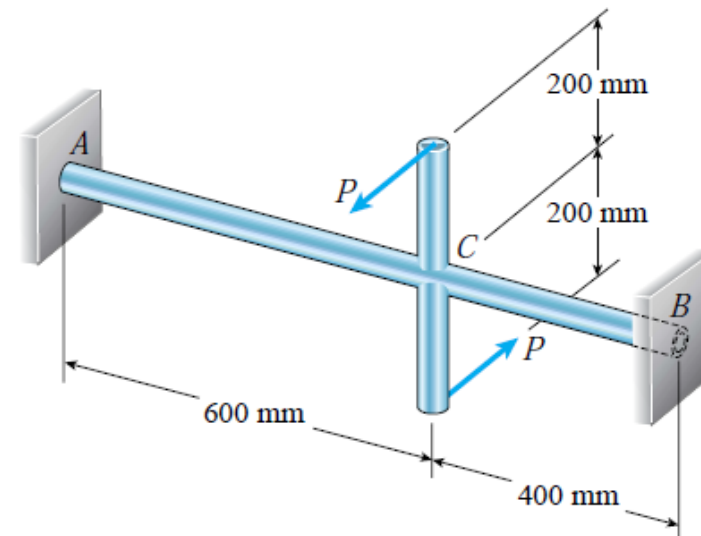
Example 3.5:

5. A solid circular bar $ABCD$ with fixed supports is acted upon by torques T_0 and $2T_0$ at the locations shown in the figure. Obtain a formula for the reactive torques T_A and T_D at the ends.



Example 3.6:

6. A hollow steel shaft ACB of outside diameter 50 mm and inside diameter 40 mm is held against rotation at ends A and B (see figure). Horizontal forces P are applied at the ends of a vertical arm that is welded to the shaft at point C . Determine the allowable value of the forces P if the maximum permissible shear stress in the shaft is 45 MPa.



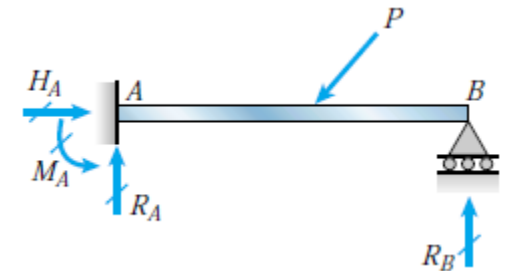
3.4 Statically indeterminate beams

Types of Statically Indeterminate Beams

Statically indeterminate beams are usually identified by the arrangement of their supports as:

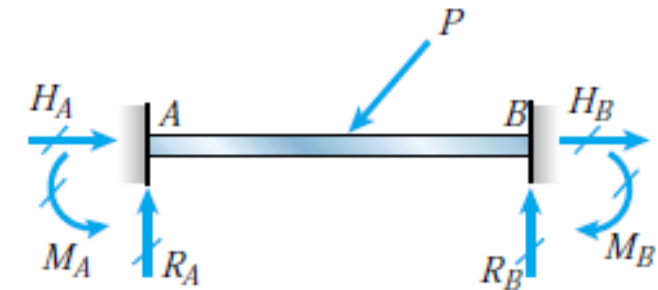
(a) Propped cantilever beam

- a beam that is fixed at one end and simply supported at the other



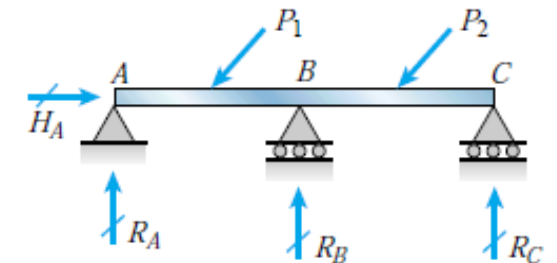
(b) Fixed end beam

- has fixed supports at both ends



(c) Continuous beam

- has more than one span and is continuous over an interior support.



Method of analyzing Statically Indeterminate Beams

Three common methods are used in solving statically indeterminate beams

- a) Using **differential equations** of the deflection curve
- b) Method of **super position** (Force or Flexibility Method)
- c) **Moment -Area** Method

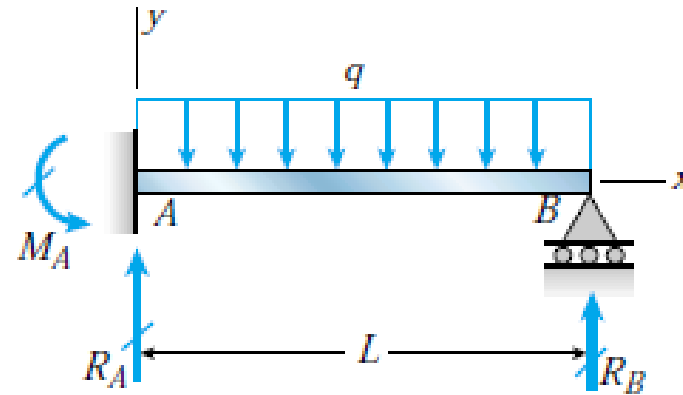
a) Using differential equations of the deflection curve

Procedures:

1. Using free body diagram write the equilibrium equation and express it in terms of the redundant reactions.
2. Express the bending moment $M(x)$ at any given point of length L of the beam in terms of the distance x , the given load, and the unknown reactions.
3. Integrate in x and obtain expressions for angular rotation, θ and deflection, y which contain two additional unknowns, namely the constants of integration C_1 and C_2
4. Apply boundary and other conditions to evaluate the unknown quantities. The unknowns consist of the redundant reactions as well as the constants of integration.

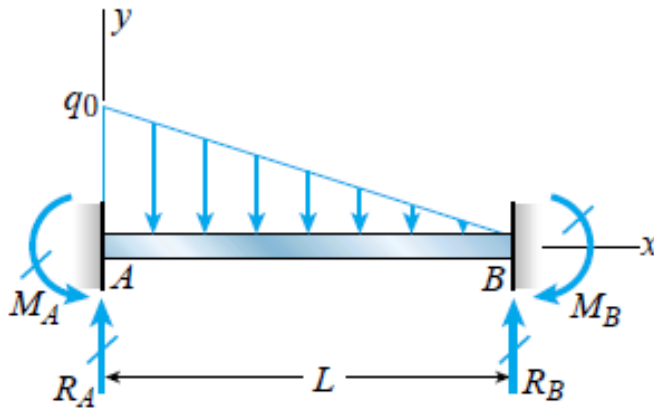
Example 3.7:

7. A propped cantilever beam AB of length L supports a uniform load of intensity q . Determine all the support reactions.



Example 3.8:

8. A fixed-end beam AB of length L supports a triangularly distributed load of maximum intensity q_0 (see figure). Using method of integration obtain the reactions of the beam at the supports.



b) Method of superposition (Force or Flexibility Method)

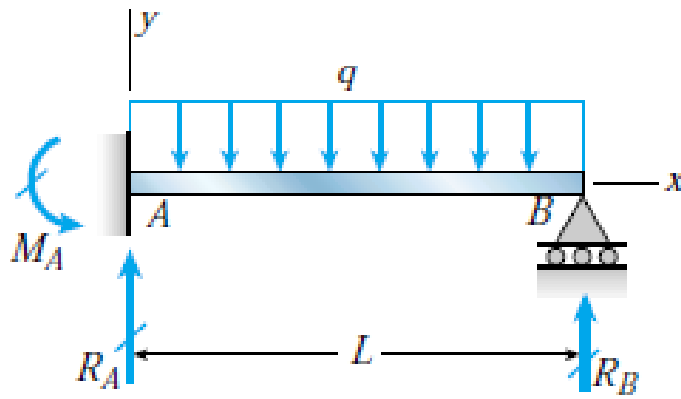
Procedures:

- Identify the degree of static indeterminacy and select the redundant reactions.
- write equations of equilibrium that relate the other unknown reactions to the redundants and the loads.
- Next, by assuming that both the original loads and the redundants act upon the released structure, find the deflections in the released structure by superposing the separate deflections due to the loads and the redundants. The sum of these deflections must match the deflections in the original beam. However, the deflections in the original beam (at the points where restraints were removed) are either zero or have known values.

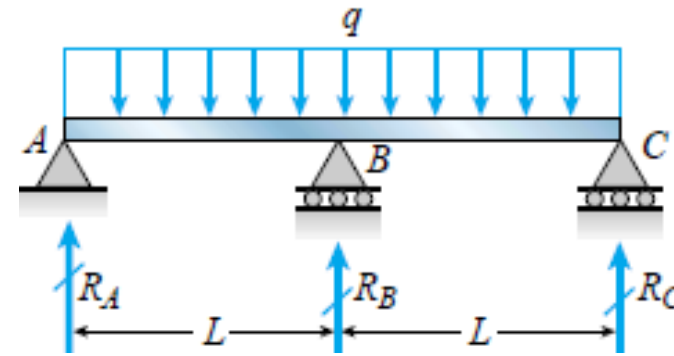
- Finally write equations of compatibility (or equations of superposition) expressing the fact that the deflections of the released structure (at the points where restraints were removed) are the same as the deflections in the original beam (at those same points).

Example 3.9:

9. A two-span continuous beam ABC supports a uniform load of intensity q , as shown in Figure. Each span of the beam has length L . Using the method of superposition, determine all reactions for this beam.



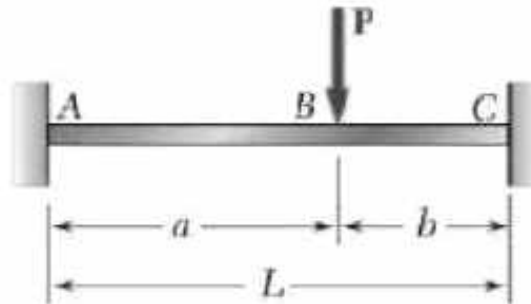
A)



B)

Example 3.10:

10. For the beam and loading shown, determine the reaction at the fixed support C using superposition method.



c) Moment Area Method

Example 3.11:

11. Determine the reaction at the supports for the prismatic beam and loading shown using moment-area method

